

Stock Liquidity and Firm Performance: Evidence from a Policy Experiment in India

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ABSTRACT

The Security and Exchange Board of India (SEBI) - the Indian securities market regulator - under the direction of Ministry of Finance, Government of India, intervened with a regulatory amendment in the listing requirement which mandated all the listed firms other than PSUs to have a minimum public shareholding of 25%. The minimum public shareholding for Public Sector Units (PSUs) was set at 10%. The affected firms were given a three year window, starting 4 June 2010, to comply with the regulation. The present research work examines the impact of the new regulation on the affected firms' performance/value. It further explores the relationship between improvement in firms' performance and key liquidity and ownership variables. It also uses this natural shock to stock liquidity to establish direct causality between stock liquidity and firms' performance in terms of stock value. The results of the ordinary least square regression analysis indicate that change in firms' performance has a positive relationship with the change in stock liquidity and promoters' block holding concentration, whereas there is a negative relationship with the change in promoters' equity dilution fraction and *ex-ante* firm size. Firms that witnessed a significant increase in the participation of institutional investors and FIIs show a positive relationship, though statistically insignificant. A DiD regression analysis shows that firms which adhered to the MPS regulation witnessed a significant and positive change in performance i.e., the industry-adjusted Tobin's Q of treatment group firm increased by 0.84 points more than the control group firms during the financial years 2010-11 to 2014-15. On breaking down Tobin's Q into sub-components namely, Operating Income-to-Price (OIP), financial Leverage (LEV) and Operating Income-to-Assets (OIOA), it was observed that the underlying mechanism of improvement was operating performance based (OIOA). MPS regulation was considered a natural shock to stock liquidity to confirm the existence of a direct causal relationship between stock liquidity and firm performance in the context of the Indian capital markets.

Keywords: Public Float, Stock Liquidity, Firm Performance, Financial market regulation

1. Introduction

Ownership structure plays a pivotal role in corporate finance decisions, governance and performance of firms, especially in the context of emerging markets. Unlike developed countries where the central theme is to align the interests of principal and agent, emerging markets witness a conflict of interest between the insider (dominant) and the outsider (small and dispersed) shareholders (LaPorta et al., 2000). A close examination of emerging market economies reveals that the firms are mostly run by business groups or families who enjoy significant control over the firm's decision making – as owners and managers of the firms. Extant research also highlights that the emerging markets are underdeveloped with very high degree of information asymmetry, providing such owner-managers a great deal of independence to make decisions based on their whims and desires.

Managerial alignment theory suggests that high ownership concentration is beneficial in mitigating the conflict between insider and outsider shareholders. However, supporters of the entrenchment theory have a diverse view, especially when it is in the context of emerging market economies having issues pertaining to quality of regulation and enforcement of contracts. A higher information asymmetry, underdeveloped and inefficient markets coupled with weak regulatory regime, create a very conducive environment for such controlling owner-managers to easily expropriate the wealth of outside shareholders and to indulge themselves in non-profit activities at the firm's expense.

Regulators from many emerging markets had been striving hard to mitigate some of the country-specific market imperfections by amending their regulatory norms. In a similar pursuit, the securities market regulator in India (SEBI) amended the listing agreement for both listed and to be listed firms on Indian bourses. The new law mandated the firms to maintain a minimum public shareholding of 25%¹. The intent was to enhance the minority shareholders' rights and interest by improving liquidity and governance in the high ownership firms. The regulation affected 286 listed firms which, on an average, had to dilute 8.8% of their promoter stake in a three-year window by choosing various methods of dilution made available to them by the regulator. This created an

¹ This was later set at 10% for Public Sector Undertakings (government owned large firms).

exogenous shock to the ownership structure of the firms as the new ownership composition was thrust upon by the regulator and had nothing to do with the changes in firm-level performance indicators. However, one thing was very evident: the huge flow of stocks into the free float market capitalization of the firms would significantly impact their liquidity.

Stock liquidity is very vital for smooth functioning of the secondary market. Researchers, with the help a host of causative theories², have shown that stock liquidity impacts positively on firm performance. However, the verdict that improved stock liquidity leads to better firm performance has not always shown unanimity, as many competing theories³ suggest otherwise and have discovered either a negative relationship or no relationship between these two constructs.

Prior theoretical works indicate that block formation of outside shareholders is a function of stock liquidity – higher the stock liquidity, better are the chances of block formation. Studies, both theoretical and empirical, have shown that such block holders improve firms' governance by means of actively monitoring managerial decisions and taking part in discussions with the management in cases of discontent, known commonly as activism (Holmstrom and Tirole, 1993; Maug, 1998; Norli et al., 2015). However, some studies (Back et al., 2013; Bhidé 1993, Coffee, 1991; Roosenboom et al., 2013) found that liquidity does not always favor activism. They argue that liquidity not only facilitates block formation but also helps in block holders' exit which acts as impedance to better governance since monitoring and activism take back seat in such cases.

Recent studies (Admati and Pfleiderer, 2009; Bharath et al., 2013; Edmans, 2009; Edmans et al., 2013; Fos and Kahn, 2015) have shown that it is not only activism that helps to improve firms' governance but exit may also prove to be a vital tool in disciplining the management. He argues that better stock liquidity not only facilitates block formation but also makes exit less costly. The institutional investors, who are supposedly better informed, trade early on learning about the managements' indulgence in non-value maximizing decisions; this is mimicked

² Agency based (Admati and Pfleiderer, 2009; Edmans, 2009; Holmstrom and Tirole, 1993; Maug, 1998), Information based – positive feedback effect (Khanna and Sonti, 2004; Subrahmanyam and Titman, 2001); Pricing based (Holmstrom and Tirole, 2001); and Perception (behavioral) based (Baker and Stein, 2004)

³ Negative Feedback, Agency Based (Bhide, 1993; Coffee, 1991); short-selling (Goldstein and Guembel, 2008) etc

by the less informed players in the market and eventually causes a drastic fall in equity prices. This reduction in firms' value impacts the owner-managers or professional managers if their incentive scheme is tied up with the equity option. Therefore, under threat of such a crash in stock prices, managers refrain from taking non-value maximizing decisions and run the firm to the best of their ability, which leads to better governance and performance.

Another school of thought offers a price based feedback theory, which says that stocks are traded more as liquidity improves. Prices of stocks that change more hands depict true value better. Thus any wrong or non-value maximizing decisions by the firms would reflect in the stock prices sooner; therefore managements would like to correct their decisions in order to reverse any such negative price movements, especially in cases where their incentive is tied to equity options.

A detailed review of literature indicates that there is theoretical and empirical evidence, in favor as well as disfavor, of this causative relationship (Cheung et al., 2015; Fang et al., 2009; Fang et al., 2015; Nguyen et al., 2016; Wu and Liu 2011). Most of these studies have been conducted in the context of a developed market where both agency- and information-based theories are of very different nature compared to those of emerging market economies. This void creates an opportunity to explore the relationship in the context of this unique emerging market where the agency problem majorly originates from Principal-Principal conflict, with a very high degree of information asymmetry in a weaker legal regime. Moreover, almost all these causative studies have been conducted in respect of exogenous shock on liquidity for the decimalization era of 2001 in the US markets. Of late, researchers have questioned the unbiased treatment and control group formation for such causative studies (Back et al., 2013) using decimalization. Also, drawing any decisive conclusion from a one and a half decade old event may have its own biases on the applicability of the underlined theories in today's market environment. Last but not the least, to the best of our knowledge, no empirical study has been made till date to examine the impact of such a regulatory change⁴ on firms' performance. Moreover, many emerging market economies in Asia, in pursuit of a healthy secondary market, are mimicking the regulatory move in some form or the other, and timely

⁴ Increase or decrease of public shareholding (free-float) requirement for continued listing, like MPS in India.

research on the impact of such a regulatory move could be of interest to the stakeholders and policy makers from the emerging market world.

With a view to contribute incrementally and fill some of the gaps identified in the extant literature, this paper has three key objectives namely,

- (i) to understand the relationship between change in firms' performance with a change in stock liquidity and ownership structure;
- (ii) to study the impact of MPS regulation, as a whole, on the firms' performance and to explore the causal mechanism of the same (operating performance based or price based); and,
- (iii) to retest the causal relationship between stock liquidity and firms' performance, especially in the Indian context .

The standard Ordinary Least Square (OLS) regression analysis on liquidity, ownership and performance measures of the affected firms is undertaken to understand the relationship between changes in these variables before and after the imposition of MPS regulation. Further, a Difference-in-Difference regression analysis has been done on various control and treatment groups of firms, to understand the impact of MPS regulation on firms' performance and to identify its causal mechanism. We conduct another Difference-in-Difference regression analysis on two sets of affected firms, with positive and negative changes in liquidity, to test the direct causal effect of liquidity on firms' performance.

The key findings indicate that the revised MPS regulation, on the whole, had a positive impact on firms' performance. Firms that had a positive change in liquidity after the imposition of the revised MPS regulation also witnessed significant higher firms' value compared to those firms that had no change or negative change in liquidity. This establishes a causal relationship between stock liquidity and firms' performance.

OLS regression on change of key performance in ownership and liquidity variables reveals that decrease in ownership level, concentration of insiders (promoters) and increase in block holding of institutional investors cause significant positive impact on firms' value post-regulation . Increase in the level of institutional ownership,

however, was found to be weakly significant in explaining the post-regulation value of firms. It was also found that improvements in stock liquidity (price based liquidity measures) significantly affected the post-regulation value of firms positively.

This research work contributes to the body of knowledge related to the role of liquidity in firms' performance in an incremental manner. To the best of our knowledge, this research work is the first of its kind to study the impact of such a regulatory intervention by securities market regulator on firms' performance. The study further throws light on the effects of change in ownership structure and stock liquidity on firms' performance. The study also successfully establishes the causal relationship between stock liquidity and firms' performance, in the Indian context.

2. Literature Review and Hypotheses Development

There are several studies emanating from different causative theories that indicate that stock liquidity has a significant relationship with firms' value/performance. On the positive association between stock liquidity and firms' performance, agency based causative theories such as, the one given by Holmstrom and Tirole (1993) and Maug (1998) and others, suggested that liquid stock market helps block formation, which in turn helps in better intervention and monitoring, resulting in better governance and firms' performance. On similar lines, Holmstrom and Tirole (1993) suggested that stock liquidity promotes efficient managerial compensation while Edmans (2009), Admati and Pfleiderer (2009), Palmiter (2001) etc., suggested that liquidity in stock reduces managerial opportunism. Subrahmanyam and Titman (2001), and Khanna and Sonti (2004), argue that liquidity stimulates more trade by informed players in a stock, thereby improving price discovery and hence disciplines management by facilitating decision making which positively impacts firms' performance. This is commonly known as information based causative theory of positive feedback. The next strand of causative theory emanates from pricing based arguments that speak of the cost of illiquidity – known as illiquidity premiums or illiquidity mispricing (Amihud & Mendelson, 2008). Since the liquid stocks trade at a premium, they have better stock prices compared to similarly performing illiquid firms, and hence they command a better market value (Holmstrom and

Tirole, 2001). Apart from these, some studies also show that liquidity is associated with better sentiments which in turn, improve stock prices and hence firms' market value (Baker and Stein, 2004).

On the other hand, there are theories that contradict the positive relationship between stock liquidity and firms' performance. Coffee (1991) and Bhidé (1993) argue that stock liquidity makes the exit of block holders easier and in the case of management's deviation from value maximizing decisions, they prefer exit over costly monitoring – known as “voting with their feet” – resulting in suboptimal managerial decisions and decline in firm value/performance. In an empirical study, Gaspar and Massa (2007) show that there exists a tradeoff between monitoring by block holders and stock liquidity. They argue that informed block holders improve firms' performance through better governance and value-enhancing decisions by the management. However, such block formation, in turn, reduces stock liquidity, which increases the adverse selection cost by increasing the information asymmetry of the stock, and hence has a decreasing effect on the firms' value.

With regard to information based causative theories on negative feedback effect, Goldstein and Guembel (2008) have proved empirically that higher liquidity of a stock is prone to exploitation by speculative traders because of various short-selling strategies that negatively impact firms' performance.

Thus to conclude, causative theories on the relationship between stock liquidity and firms' performance are based either on operating performance or pricing. We could segregate the operating performance based theories into two main categories – agency based and information based.

No matter what the causative mechanism may be, governance based performance improvement or price based feedback effect, one may conclude that literature is more biased towards the positive relationship between liquidity and firms' performance, though the relationship has no unanimous acceptance till date. Also, changes in ownership composition in tandem with liquidity have implications on firms' performance. There is a common observation based on the findings of many researchers that institutional investors undertake the role of monitors and help in making the management more disciplined which, in turn, improves the firms' performance via improved governance (Chen et al., 2007; Chung et al., 2002; Hartzell and Starks, 2003; McConnell and Servaes,

1990). In the backdrop of the above-cited literature and arguments, we propose the following hypotheses on the tradeoff amongst stock liquidity, ownership structure and firms' performance:

- *H1: There is a significant positive relationship between change in stock liquidity and firms' performance.*
- *H2a: There exists a significant positive relationship between increase in outsider (non-promoter) block holders' concentration and firms' performance.*
- *H2b: There exists a significant positive relationship between increase in insider (promoter) block holders' concentration and firms' performance.*
- *H3a: There exists a significant positive relationship between change in institutional ownership level and firm performance.*
- *H3b: There exists a significant negative relationship between change in insiders' ownership level and firm performance.*

Out of the many classes of investors, it is the foreign institutional investors (FIIs) who pressurize the management to disclose quality information timely (Gul et al., 2010). It has also been empirically found that presence of FIIs in a company's shareholding pattern attracts more analyst coverage which, in turn, automatically becomes an additional monitoring force that helps make the managers more disciplined. They also help reduce information asymmetry about the quality of management and the decisions taken (Bae et al., 2006). Wei et al. (2005) in their study find that FIIs are more sophisticated and have superior managerial skills that help them to restrain managers from indulging in any insider trading or diverting company's resources for private benefit.

On the other hand, FIIs, in particular, have always been considered to be more active than domestic players in churning portfolios, which has serious implications on managerial decision-making, called the problem of managerial myopia. Also, according to the negative effect of price based feedback theories, it is observed that FIIs are more engaged in speculative trading and short selling, which further hamper firms' performance. Thus, it is evident from the above literature that the role of FIIs in firms' performance is still inconclusive and debatable. However, nature of the relationship is uncertain in the aforesaid context. We expect a significant relationship of

FII holdings on the firms' performance. In the backdrop of the above-cited literature and arguments, we propose the following hypothesis:

- *H4: There exists a significant positive relationship between change in Foreign Institutional Investors' holding (FIIs holding) and firms' performance.*

The MPS regulation had a multidimensional impact on firms' information and governance attributes such as, reduced promoter control, equity dispersion and improved institutional participation. The regulation also helped in increased block holding concentration of promoters and non-promoters. These changes, in tandem with improved stock liquidity, helped in reducing firms' information asymmetry which, in turn, would reflect in higher firm valuation. Irrespective of the underlying mechanism of improvement in firms' performance – be it operating performance based (through activism or through the threat to exit) or price based - the feedback effect, on the whole, should improve the firms' valuation and performance. We, therefore, propose the next hypothesis as follows:

- *H5: MPS regulation has a significant positive impact on the performance of compliant firms.*

As established by Fang et al. (2009), the measure of firm value/performance can be segregated into price based and operating performance based measures. If the underlying mechanism of improved performance of firms is operating performance based, one should expect a significant change in the operating income to asset ratio of the firms. And if the underlying mechanism is price-based, one should expect a significant improvement in the leverage and operating income to price ratios. In light of the aforesaid arguments, we propose the following set of hypotheses:

- *H6a: MPS regulation has a significant positive impact on the operating-performance-to-asset ratio of compliant firms.*
- *H6b: MPS regulation has a significant positive impact on the operating performance to price ratio of compliant firms.*
- *H6c: MPS regulation has a significant positive impact on the leverage ratio of compliant firms.*

Although it is theoretically established from the decade old liquidity shock experiment (decimalization in the US) that stock liquidity helps improve firms' valuation, there has been disagreement, however, in the causative mechanism, as we find empirical evidence in support of both. Moreover, a recent study by Nguyen et al. (2016) shows that the liquidity-performance relationship holds true for Australian equity market. They found that this causal relationship was price based whereas Fang et al. (2009) found it to be operating performance based. Thus it is evident that the debate is yet quite inconclusive. Further, since emerging market economies face a very different nature of institutional set-up, it would be worthwhile to test this relationship in the context of the Indian market. Chung et al. (2010) investigated this relationship from the viewpoint of reverse causality and argued that in spite of liquidity attracting institutional participation, high Tobin's Q firms might be attracting liquidity traders. In the light of the above argument, the next hypothesis is proposed as under:

- *H7: There is a direct causal relationship between stock liquidity and firm's value, i.e., improved stock liquidity leads to a better market valuation of the complaint firm.*

3. Research Methodology

3.1 Data

Data needed for the empirical analysis was collected from secondary sources only. We collected the information in respect of the MPS regulation from various publically available websites organizations and ministries such as, SEBI, BSE, NSE and MoF, of the Government of India. We used these sources of information to understand the institutional set-up on MPS regulation for continued listing in India and subsequently to highlight the key parameters of the revised MPS regulation such as, the timelines and prescribed methods of compliance, etc.

We fetched promoters' stake dilution data from PRIME database and cross-checked them with the data archives of NSE and BSE. The data obtained from these sources contained information on the method, size and dates/period of equity dilution, the rate of success in the process etc. We obtained quarterly shareholders' ownership pattern data from various sources like BSE data archive and PROWESS database provided by Centre for Monitoring Indian Economy Pvt Ltd. (CMIE-Prowess).

In addition to the events and ownership data, we collected trading data of the affected firms such as, stock price, trading volume, shares outstanding, and other market-based stock performance ratios. We obtained firm level identity and financial characteristics data from the CMIE-Prowess database. We also collected firm-level financial data for the financial years 2009-10 to 2015-16 from the standalone annual financial data section of CMIE-Prowess database.

We calculated annualized ownership, liquidity and returns from the daily and quarterly data for 2010-11 and 2014-15, i.e., one year before the promoters' ownership dilution and one year after the end of the MPS regulation adherence window deadline.

In order to conduct the Difference-in-Difference regression analysis for analyzing the overall impact of MPS regulation on the firms' performance/value, we constructed a matched firms' portfolio based on matching industry and firm characteristics. We also classified treatment group firms based on their overall liquidity improvement matrix into three different groups – positive, neutral and negative. We then considered firms from the positive liquidity change group as the treatment group and those with negative change in liquidity as the control group for the causality test of liquidity on firm performance/value.

3.2 Empirical Framework, Variables and Sample Selection

We used simple OLS regression to test the relationship between change in ownership composition and stock liquidity on firms' performance caused by the MPS regulation. We calculated the change in dependent and independent variables from pre-regulation to post-regulation period i.e., between the financial years 2010-11 and 2014-15. The model specifications are as follows: (variable definitions and methods of calculation are outlined in Table 1)

$$\text{Model 1: } \Delta \text{ Tobin's } Q_IA = f(\Delta \text{Amihud_Illiq}, \Delta \text{Tr_Vol})$$

$$\text{Model 2: } \Delta \text{ Tobin's } Q_IA = f(\Delta \text{Pr_Eqty_Own}, \Delta \text{Int_Inv_Own})$$

$$\text{Model 3: } \Delta \text{ Tobin's } Q_IA = f(\Delta \text{Pr_Eqty_Own}, \Delta \text{FII_Own})$$

$$\text{Model 4: } \Delta \text{ Tobin's } Q_IA = f(\Delta \text{Pr_Eqty_Own}, \Delta \text{Int_Inv_Own}, \Delta \text{FII_Own})$$

Model 5: $\Delta \text{Tobin's } Q_{IA} = f(\Delta HHI_Pr, \Delta HHI_Non-Pr)$

Model 6: $\Delta \text{Tobin's } Q_{IA} = f(\Delta Amihud_Illiq, \Delta Tr_Vol, \Delta Pr_Eqty_Own, \Delta Int_Inv_Own, \Delta HHI_Pr, \Delta HHI_Non-Pr)$

Model 7: $\Delta \text{Tobin's } Q_{IA} = f(\Delta Amihud_Illiq, \Delta Tr_Vol, \Delta Pr_Eqty_Own, \Delta FII_Own, \Delta HHI_Pr, \Delta HHI_Non-Pr)$

Model 8: $\Delta \text{Tobin's } Q_{IA} = f(\Delta Amihud_Illiq, \Delta Tr_Vol, \Delta Pr_Eqty_Own, \Delta Int_Inv_Own, \Delta FII_Own, \Delta HHI_Pr, \Delta HHI_Non-Pr)$

In order to understand the mechanism which explains the improvement or deterioration in firms' performance due to change in stock liquidity, we separated Tobin's Q into the following three components using the method specified by Fang et al. (2009):

- Operating Income to Price Ratio (OIP);
- Financial Leverage (LEVERAGE); and,
- Operating Income to Assets ratio (OIOA).

The suggested relationship, as outlined in the works of Fang et al. (2009), is as under:

$$\text{Tobin's } Q_{it} = (1/OIP_{it}) \times (1/Leverage_{it}) \times OIOA_{it}$$

The key variable of interest for this analysis is firm performance. We calculated Tobin's Q, the most commonly used measure of firms' performance in liquidity-performance studies (Fang et al., 2009 and Nguyen et al., 2016; Wu and Liu, 2011). Then firm performance was split into three components, as suggested by Fang et al. (2009) and Nguyen et al. (2016), into OIP, Leverage and OIOA. To ascertain the effect of industry's performance as a whole on individual firm's performance, we adjusted the Tobin's Q and its subcomponents by subtracting annual performance measures from industry means.

The independent variables comprise of ownership level, promoters' block holding concentration, stock liquidity - price and volume based⁵. We also included several control variables based on the directions given by

⁵ Since Goyenko et al. (2009) illustrated the need for using multiple measures to capture different liquidity dimensions.

prior empirical works in this domain. We considered firm size, age, and cross-listing⁶ as control variables. We also used change in debt to equity ratio of the firms and stock momentum (measured as the last one-year cumulative average market returns). Variable definitions and methods of calculation are outlined in Table 4.1 (p.155).

We then conducted Difference-in-Difference regression analysis for estimating the causality of stock liquidity on firms' performance, prior to and post liquidity shock period. For this purpose, we considered MPS regulation as one such event of exogenous shock on the stock liquidity of the affected firms. We segregated the complaint firms into treatment and control groups, based on change in their liquidity values. Top forty firms with positive change in stock liquidity from amongst the firms affected by MPS regulation were considered as the treatment group, and the bottom forty firms with highest negative change in liquidity were taken as the control group. For robustness check, we also conducted causality test on a different group of control firms that were not affected by the MPS regulation but were similar as per industry classification and firm/stock level characteristics.

We chose the Difference-in-Difference method as it has multiple advantages over other methods of causality analysis (Atanasov and Black, 2016; Roberts and Whited, 2012). This method avoids the issue of omitted trend in treatment and control group firms that have a correlation with firms' performance and liquidity over time. Also, this method mitigates the issue of unobserved differences in the characteristics of the two groups.

In this study, we have only considered MPS affected firms that complied with the regulation, and were not from the banking and finance sector. We also dropped observations that had missing data. Out of 286 affected firms, 191 firms complied with the regulation. Out of these, 61 were financial, and 225 were non-financial firms. Only firms that complied with the regulation by March 2014 were considered, reducing the sample size further.

⁶ Cross-listed at NSE, a stock exchange with stringent listing requirements. It is considered a signal of better transparency and quality in listed firms.

4. Results and Discussions

4.1 Change in Liquidity, Ownership Structure and Firm Performance

We performed stepwise regression on the change in firms' performance (Industry adjusted Tobin's Q) with a change in stock liquidity and ownership structure. Change in ownership structure indicates a change in the level and concentration of promoters' equity holding, change in the level of Institutional Investors & FIIs and change in the concentration of promoters' block holdings. Taking lead from previous studies such as, Fang et al. (2009), Prommin et al. (2016) and Nguyen et al. (2016), which also explored this relationship, we considered firms' other characteristics such as, *ex-ante* firms' age, size, returns momentum, for the previous year, cross-listing dummy, change in debt to equity ratio etc., as control variables. Summary statistics is presented in Table 2.

The results of the stepwise OLS regression analysis, as reported in Table 3, indicate that the change in liquidity is positively related to change in firms' performance. A closer look at all the four model specifications shows that every unit change in price-based measure of illiquidity (Amihud illiquidity ratio) helps improve firms' performance by -0.0381, -0.042, -0.0421 and -0.042 respectively, at $p < 0.05$, $p < 0.1$, $p < 0.1$, and $p < 0.1$ for model 1, 6, 7 and 8 respectively. Volume based liquidity measure also shows a similar trend. However, parameter estimates being econometrically insignificant, no convincing inference can be drawn. On the whole, the results are in consonance with the view expressed in extant literature (Amihud and Mendelson, 2008; Cheung et al, 2015; Fang et al., 2009; Prommin et al, 2016; Wu and Liu, 2011; etc) which argue that stock liquidity has a positive impact on firms' value/performance.

We analyze the relationship between firms' performance and ownership variables in two steps. In the first step, we regress firms' performance with the level of institutional and promoter holdings. We find that change in promoters' equity ownership is positively and significantly related to change in firms' performance. The test statistics reveal that every one percent ownership stake dilution affects firms' performance negatively by 0.0983, 0.0976 and 0.102 unit, which are econometrically significant for model 2, 3 and 4 at $p < 0.1$, $p < 0.1$ and $p < 0.1$ respectively. In the scenario under study, this means that dilution in promoter ownership has value decreasing

effect on the firms' performance, which is in line with the interest alignment theory of ownership and firm performance (e.g., Demsetz and Lehn, 1985; James, 1999; Stein, 1989). We also look at the relationship between changes in institutional investors' ownership composition and firms' performance. One percent point change in the level of Institutional investors' holdings results in improvement in firms' Tobin's Q by 0.039. The parameter estimates are however econometrically insignificant to take away anything concrete from these numbers. When we consider FIIs in place of institutional investors, we find that the relationship between changes in FIIs holdings and firms' performance is positive, which is in consonance with general understanding and belief. However, when we consider both FIIs holdings and institutional holdings in the regression equation, we find the results are still insignificant; however, we witness a negative relationship between firms' performance with FIIs holdings. The result, in this case, is however statistically insignificant, and we do not conclude anything concrete from these numbers. But, the increase in FIIs holdings' negative relationship with firms' performance hints at a possible existence of short selling behavior of the FIIs and managerial myopia that could be creeping in, which at times have value decreasing effect on the firms' performance.

We further regress firms' performance with the block holding concentrations for both promoters and non-promoters. The results indicate a significant positive change in the block holding concentration within the promoter class of shareholders. A one percent increases in promoters' ownership concentration helps increase firms' performance (Tobin's Q) by 0.022, which is econometrically significant at 1% level. Therefore, it is found that dispersed *ex-ante* promoters' ownership, which is getting concentrated after the regulation, had a positive and significant impact on the firms' performance/value. Improvement in the firms' performance might be due to increased focus and reduced free-rider problem within firms' management, which might have helped rejuvenate the managerial focus in firms' value maximization or in creating such a perception in the marketplace. On the other hand, this would make the promoters more powerful within their respective boards and less contested than before in taking decisions in the board meetings, taking a long-term perspective.

We then regress firms' performance with all the key liquidity and ownership variables along with the most commonly used control variables in literature on firms' performance and liquidity. We present the coefficient

estimates and the test statistics in columns 6, 7 and 8 of Table 3. The results indicate that price-based liquidity measure, along with promoter ownership and block holding concentration, impact change in firms' performance significantly at 10%, 5%, and 5% levels respectively. We also observe that ex-ante firm size has a negative relationship with change in firms' performance. Thus, smaller firms witnessed higher improvement in firm performance as compared to the large firms in the sample. We also observed that firms that were cross-listed at NSE witnessed higher improvement in performance as compared to those who were listed only on BSE.

To conclude, the results discussed as above, on the whole, are in consonance with the hypotheses developed for the relationship between liquidity and ownership variables with firms' performance. The detailed results are given in Table 3.

4.2 Impact of MPS Regulation on Firm Performance/Value

The impact of MPS regulation on firms' performance/value is tested in this section. Treating the firms that had to dilute promoter stake due to MPS regulation as treatment groups, we compare the difference in their performance before and after the regulation with respect to the change in firm performance of the chosen control group firms, which are very similar in characteristics. The results of the Difference-in-Difference regression analysis, as shown in column 1 of Table 4, indicate that post regulation, there is a positive change (significant at $p < 0.01$) in firms' performance (value) measured as industry-adjusted Tobin's Q.

Following Fang et al. (2009) and Nguyen et al. (2016), Difference-in-Difference analysis was performed by replacing Tobin's Q with its sub-components, namely OIP, Leverage and OIOA. The results, as shown in columns 2, 3 and 4 respectively of Table 4, indicate that the regulation had a positive impact on all the three sub-components of firms' performance though pricing-based measures namely, OIP and Financial Leverage, were econometrically insignificant. Operating performance based measure (OIOA), however, showed an econometrically significant positive change (at $p < 0.1$) due to the regulation when compared with similar control firms, not subjected to the MPS regulation. These outcomes establish that firms affected by the MPS regulation were better managed after promoters' equity dilution. The causal mechanism is not well established. This study,

however, concludes that it is not the liquidity premium which makes an impact on firms' performance. The firms are better managed and governed due to either activism or the threat of exit of institutional investors. Apart from the aforesaid causes of improved firms' performance, promoters' block holding concentration might also be a potential cause in consonance with the managerial alignment theory.

4.3 Causality - Stock Liquidity to Firm Performance

In order to establish the causality between stock liquidity and firms' performance, an exogenous liquidity shock is needed. The MPS regulation provided one such shock to stock liquidity, which came as a blanket regulatory change and had no firm-specific information effect. If one could control the change in ownership parameters, it would then be easier to test the causality of stock liquidity to firms' performance. We, therefore, arranged the regulation compliant firms into three subgroups, based on the effect on liquidity measures. We calculated the equal weighted average annual change in firms' stock liquidity for the regulation period 2010 and 2014. Based on the final score, we arranged the firms into three sub-groups. The first sub-group comprised firms that witnessed significant positive change in stock liquidity. The second sub-group firms had marginal or no impact of regulation on stock liquidity. Lastly, the third sub-group of firms witnessed significant negative change in stock liquidity. We then chose the first and third sub-group firms as treatment and control groups respectively for testing the causality effect using Difference-in-Difference regression analysis.

We took both treatment and control group firms from the same pool of MPS affected firms in order to control change in ownership structure due to MPS regulation. Literature suggests that ownership structure affects firms' performance in one way or the other. We did the Difference-in-Difference regression analysis on firms' Tobin's Q for values of the financial year 2010-11 and 2014-15. Results of the analysis are presented in column 1 of Table 5. We find that positive liquidity shock improves firms' performance positively. The results are econometrically significant at 10% level.

As a robustness test, we controlled industry effect by adjusting Tobin's Q values with the industry mean values taken from industry classification data available on CMIE Prowess Industry Outlook database. We then

conducted the Difference-in-Difference regression analysis. The results are presented in column 2, Table 5. The results are found to be in line with the initial findings and are econometrically significant at 10% level.

We, therefore, conclude that there exists a direct causal relationship between stock liquidity and firms' performance in an emerging capital market like India.

5. Conclusion

In this research work, we have examined the relationship between stock liquidity, ownership structure and firms' performance, under the influence of exogenous natural shock to ownership and stock liquidity, caused due to the change in MPS regulation in India. We find that price-based measure of stock liquidity, percentage dilution of promoters' ownership and promoters' block holding concentration are significantly as well as positively related to firms' performance. The results hold true after adjusting firms' performance with the respective industry means. After splitting the data based on median firm size (Total Assets), we found that the model explains this relationship for small firms more than large firms.

In order to understand the underlying mechanism of improved performance of firms, we further analyzed the impact of MPS regulation on firms' performance using Difference-in-Difference regression analysis on firms' Tobin's Q and its sub-components namely, OIP, Financial Leverage and OIOA. The results indicate that the treatment firms (those affected by the MPS regulation) show significant improvements in firms' performance when compared with the control group firms (firms with similar characteristics but not affected by MPS regulation). When regressed with the sub-components of Tobin's Q, there is a positive impact on all the sub-components, but only OIOA shows econometrically significant positive change. The results are found to be robust when examined for industry adjusted firms' performance and its sub-components. We, therefore, conclude that the MPS regulation did significantly and positively impact complaint firms' performance and the underlying mechanism was operating performance based.

Lastly, we used the MPS regulation as an exogenous natural liquidity shock and observed that there existed a causal relationship between stock liquidity and firms' performance (Tobin's Q). The results were

validated for industry-adjusted values of firm performance. We also tested the relationship by controlling the variables related to firms' level of ownership structure and found the results to be in agreement with the key findings of the study, though econometrically weakly significant.

The results of the analysis have wider managerial and policy related implications on capital market regulators in the emerging markets. The results of the policy shift show improvements in the overall performance of firms' (value), which is in the larger interest of minority shareholders. At the firm level, the study shows that liquidity management by firms, in the long run, would help improve firms' value in the market.

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Table 1
Variable Definitions

Variables	Definitions
Firm Performance	
Tobin's Q	Tobin's Q is computed as the market value of equity plus book value of total debt divided by book value of total assets. $(\text{Market Capitalization} + (\text{Total Asset} - \text{Total Equity})) / \text{Total Asset}$
Tobin's Q_IA	Industry adjusted (mean-adjusted) value of the Tobin's Q
OIP	(Operating income) / (Market Value of the Common Equity), which is computed as,
Leverage	(Market Value of Equity) / (Market Value of Debt)
OIOA	EBTDIA _{it} / Total Asset
Ownership Levels	
Pr_Eq_Own	Percentage equity holdings of Promoters
Pr_Dilution	Fraction of Promoters' equity stake diluted of the total firm's equity holding
Int_Inv_Own	Percentage equity holdings of all institutional investors
FII_Own	Percentage holding of Foreign Institutional Investors
Ownership Concentration (Block Holding)	
HHI_Pr	Block holder concentration in within firms' Insiders (promoters), measured as HHI index of key reported promoters in CMIE prowess ownership database
HHI_Non-Pr	Concentration of Block holders, other than Promoters (Non-Insiders), measured as HHI index of key reported non-promoters in CMIE prowess
Stock Liquidity	
Amihud_Illiq	Daily ratio of absolute stock return to trading volume in Indian rupees, averaged for one year period and scaled up by 10,00,000 (Amihud, 2002)
Tr_Vol	Average annual trading volume
Control Variables	
ln_Size	Natural logarithm of the firm's total assets in millions of Indian rupees
ln_Age	Natural logarithm of the number of years since listing on the stock exchange
NSE_Dummy	1 if listed at NSE, else 0
Debt_Eq	Debt to Equity Ratio calculated as total debt to total asset
Momentum_2014	Past 12-month market returns

Table 2
Summary statistics

The table below presents the descriptive statistics of the sample. The definition of the variables used is the same as described in Table 1. Δ denotes the change in values of those variables – before and after the MPS regulation (difference of 2014-15 2010-11).

Variable	Mean	Std. Dev.	Min	Max
Δ TobinsQ	0.65	3.59	-5.74	25.45
Δ TobinsQ_IA	-0.07	3.4	-5.88	25.31
Δ Amihud_Illiq	4.9	14.22	-19.71	115.64
Δ Tr_Vol	-13847.06	261278	-1077231	1511473
Δ Int_Inv_Own	1.81	5.94	-21.82	21.306
Δ FII_Own	2.14	5.13	-12.66	19.44
Δ Pr_Dilution	8.4	6.24	0	-25
Δ HHI_Pr	15.83	40.37	-134.93	229.05
Δ HHI_Non-Pr	-2.38	86.41	-305.43	270.5
ln_Size(bf)	7.9	2.32	1.59	12.76
ln_Age(bf)	3.52	0.55	1.61	4.75
NSE_Dummy	0.51	0.5	0	1
Momentum_2014	-0.002	0.006	-0.023	0.034
Δ Debt_Eq	126.71	625.78	-100	5624.39

Table 3**Results of OLS regression of change in firms' performance with liquidity and ownership structure**

The table below shows the result of stepwise OLS regression between Tobin's Q (dependent variable) and Liquidity ownership and control variables using models 1 to 8 as described in section 3.2. Values in the parenthesis are the t-statistics with *p<0.1, **p<0.05, ***p<0.01

	(1) Δ TobinsQ	(2) Δ TobinsQ	(3) Δ TobinsQ	(4) Δ TobinsQ	(5) Δ TobinsQ	(6) Δ TobinsQ	(7) Δ TobinsQ	(8) Δ TobinsQ
ΔAmihud_Illiq	-0.0381** (-2.24)					-0.042* (-1.80)	-0.0421* (-1.80)	-0.042* (-1.78)
ΔTr_Vol	6.16E-7 (0.77)					1.43E-6 (1.48)	1.43E-6 (1.48)	1.43E-6 (1.46)
ln_Size	-0.691*** (-3.99)	-0.705*** (-3.69)	-0.703*** (-3.65)	-0.709*** (-3.66)	-0.603*** (-3.32)	-0.872*** (-4.03)	-0.865*** (-3.93)	-0.867*** (-3.92)
ln_Age	-0.229 (-0.49)	-0.281 (-0.52)	-0.287 (-0.52)	-0.323 (-0.58)	-0.174 (-0.37)	-0.346 (-0.65)	-0.313 (-0.59)	-0.354 (-0.66)
NSE_dummy	1.002 (1.22)	0.94 (1.01)	0.918 (0.98)	0.879 (0.93)	1.62* (1.89)	1.68* (1.74)	1.73* (1.78)	1.679* (1.71)
Mom_2014	-42.58 (-1.02)	-89.3 (-1.54)	-89.14 (-1.53)	-89.17 (-1.52)	-47.65 (-1.14)	-46.19 (-0.78)	-46.16 (-0.78)	-45.67 (-0.77)
ΔDebt_Eq	-5.88 E-6 (-0.01)	-1.35 E-3 (-0.86)	-1.41E-3 (-0.89)	-1.37 E-3 (-0.86)	-5.6 E-4 (-0.80)	3.7E-4 (0.19)	3.49E-4 (0.18)	3.75E-4 (0.19)
ΔPr_Eqty_Own		0.0983* (1.74)	0.0976* (1.72)	0.102* (1.78)		0.126** (2.19)	0.122** (2.12)	0.126** (2.17)
ΔInt_Inv_Own		0.039 (0.67)		0.062 (0.66)		0.053 (0.91)		0.063 (0.69)
ΔFII_Own			0.028 (0.35)	-0.038 (-0.30)			0.053 (0.6)	-0.019 (-0.14)
ΔHHI_Pr					0.022*** (2.64)	0.023** (2.54)	0.023** (2.48)	0.023** (2.51)
ΔHHI_Non-Pr					0.002 (0.52)	-0.004 (-0.79)	-0.003 (-0.73)	-0.004 (-0.79)
_cons	6.063*** (2.82)	6.841*** (2.71)	6.915*** (2.72)	7.132*** (2.77)	4.425** (2.07)	8.309*** (3.12)	8.099*** (3.05)	8.308*** (3.1)
N	100	86	85	85	94	81	81	81
R-sq	0.209	0.224	0.226	0.23	0.244	0.353	0.349	0.353
Adj. R-sq	0.149	0.155	0.155	0.149	0.182	0.25	0.245	0.239

Table 4
Impact of MPS regulation on firms' performance and its sub-components

The table below reports the Difference-in-Difference regression analysis conducted using the model Industry – Adjusted (Tobin's Q_i or OIP or LEV or OIOA) = $\beta_0 + \beta_1 \cdot \text{period}_i + \beta_2 \cdot \text{treated}_i + \beta_3 \cdot \text{period}_i \cdot \text{treated}_i + \epsilon_i$. Columns (1-4) report the test statistics for Industry-Adjusted Tobin's Q, Industry-Adjusted Operating Income-to-Price, Industry-Adjusted Financial Leverage and Industry-Adjusted Operating Income-to-Assets, respectively.

The values in parenthesis () reports the t-statistics and *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

	(1)	(2)	(3)	(4)
	Tobin's Q	OIP	LEV	OIOA
Difference-in-Difference Statistics				
Coefficient of Difference at Baseline (Treatment-Control)	0.003 (0.01)	-0.117*** (-4.21)	-0.145 (-1.14)	-0.026*** (-2.65)
Coefficient of Difference at Follow-Up (Treatment-Control)	0.846*** (3.70)	-0.121*** (-4.35)	-0.097 (-0.75)	-0.002 (-0.22)
Difference-in-Difference (Diff of Follow-Up - Diff of Baseline)	0.843*** (2.62)	-0.004 (-0.10)	0.048 (0.26)	0.024* (1.71)
R Square of the Model	0.03	0.06	0.01	0.07

Table 5

DiD regression analysis for testing direct causality between stock liquidity and firm's performance

The table below reports the Difference-in-Difference regression analysis conducted using the model (Tobin's Q_i or OIP or LEV or OIOA) = $\beta_0 + \beta_1 \cdot \text{period}_i + \beta_2 \cdot \text{treated}_i + \beta_3 \cdot \text{period}_i \cdot \text{treated}_i + \epsilon_i$). Column (1) & (2) report the coefficient estimates of differences and test statistics of firms' Tobin's Q and Industry adjusted Tobin's Q .

The values in parenthesis () reports the t-statistics and *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

	(1)	(2)
Difference-in-Difference Statistics		
Coefficient of Difference at Baseline (Treatment-Control)	0.641 (0.99)	0.918 (1.40)
Coefficient of Difference at Follow-Up (Treatment-Control)	2.173*** (3.45)	2.474*** (3.77)
Difference-in-Difference (Diff of Follow-Up - Diff of Baseline)	1.532* (1.70)	1.556* (1.68)
R Square of the Model	0.08	0.10